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October 1, 2025
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GEOG 384

Group 3 Assignment 1

Documentation

<https://neogeoweb.ca/group3/assignment1/>

To the provincial government of Bocas del Toro,

To assist in the mission of gaining more insight on the fascinating flight path patterns of the Greater Spear-Nosed Bat in Northern Panama, we decided to investigate the spatial layout of two specific bat flight paths from 2023. To better visualize this data, we created a clear storyboard for your perusal that provides an overview of the bats and their foraging habits.

Our storymap is based in Bocas del Toro, Panama. We wanted to emphasize this in our storymap by adding a line dataset which delimited the provincial borders of the country. This feature provided the reader with a more clear picture of the place where the bats reside and migrate throughout the country.

We chose to create a storymap for the paths taken by the Greater Spear-Nosed Bat during migration because this was one of the species which had a significant amount of data regarding their flight path available through journal articles and open government data (Datos Abiertos y Geoservicios - Sinia-Panamá, n.d.; O'Mara & Dechmann, 2021). This information made it possible to visualize more accurately the flight paths taken by this species of bats as information about this was available but it wasn't always presented in the most clear way. By translating this information into a storymap, it made it more visually accessible to the reader compared to text information, allowing the reader to get a better idea of the migration paths taken by this species of bats.

We saw a storymap which had a point that followed a line which showed the path taken by a person and decided to add that into our storymap. We made this decision because our story was planned to showcase the migration patterns of the Greater Spear-Nosed Bats and adding a line that can be followed enhanced the user experience, adding a dynamic way for them to follow the bats flight path. Moreover, we also gave the user the option of viewing the flight paths taken by two specific bats which allowed them to visualize the different possible paths taken by the bats and how there are many different ways for bats to reach their destination.

The topic of environment and wildlife preservation is a major issue in today's world, so we decided to add a 3D dataset of population in nearby areas to show the potential threat of human activity near bat habitats (HDX, 2024). Moreover, we wanted to be able to visualize how close bats can get to human populations. By adding this dataset to our storymap, it highlighted the threats that the bats are facing from human activity as it continues to expand towards their habitats. Even though having a heatmap of human populations on a map can accomplish the same task as 3D polygons, the latter was chosen as even if you're viewing the map zoomed out, you can still visualize the population densities spread out throughout the region and which parts of the region has the most humans living there.

Our story map creators benefited somewhat from the inclusion of GPT guidance, specifically ChatGPT. The first prompt asked was "of which of these land cover classes would balsa flowers be found in:" plus attached list of all landcover classes in Panama (SINIA, 2021). The GPT gave a justified response of which landcover classes the Balsa grows in, *Bosque latifoliado mixto secundario* (Secondary mixed broadleaf forest) and *Rastrojo y vegetación arbustiva* (stubble and shrub vegetation). We verified this by overlaying our bat flight paths with the land cover polygons which demonstrated a clear correlation to O'Mara & Dechmann's 2021 findings. Another use of GPT was again in the preparation of

the data and served as an ArcGIS refresher: “I have a shapefile feature class in arcgis with each polygon associated with details in the shapefile attribute table, including Class_name. What steps can I take to export this into a geojson file while including the attributes?”. I used this prompt to gain clear steps broken down in the way I wanted, a purpose the automation served that a human couldn’t. I followed these steps combined with my own reasoning to export the shapefile with the landcover attributes included, something I later included in the Mapbox style visible in our project. I also used this prompt to check off a requirement on the assignment, shapefile to geojson (Seiber, 2025).

As for client expectations, we were unfortunately unable to create and implement an interactive timeline bar in our storymap due to time constraints and lack of expertise. In addition, the land cover classes shown are only that of the two eligible land classes where balsa flowers are known to grow. Ideally, we could have included all land cover classes with a colour-coded symbology but there was a space limitation for creating a tile in Mapbox.

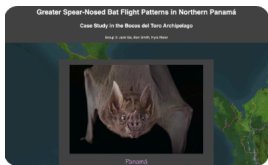
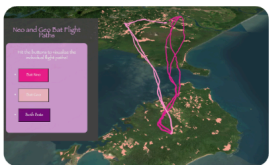
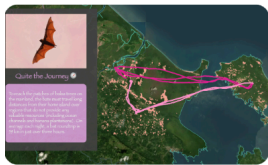
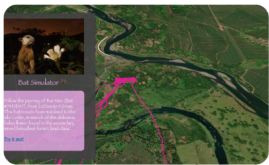
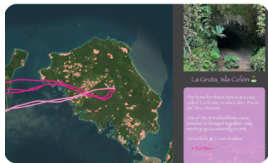
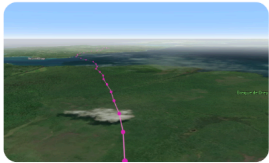
Please let us know if you have any questions regarding this product. We hope it will be useful to your provincial government and clients for the purposes of education, conservation, and familiarity with the region.

Best regards,

Kyra Meier

Ben Smith

Jack GE

	1	Title page		5	Neo and Geo Bat Flight Paths (isolation of each path possible by clicking the specified button)
	2	Overview of the Geater Spear-Nosed Bat in Northern Panamá		6	Information on La Gruta on Isla Colón (the cave habitat of the studied bats where the flight paths initiate). A video of Greater Spear-Nosed Bats in a cave is embedded for further visualization.
	3	Overview of the bat flight patterns in the Bocas del Toro Province for the 2 individuals (Bat Neo and Bat Geo). Polygons for the secondary mixed broadleaf forest land class and administrative boundary for Panamá are also displayed.		7	Bat simulator link (once pressed, an animated camera that follows the flight points for Bat Neo is opened)
	4	Information on Isla Colón (the island habitat of the studied bats where the flight paths initiate)		8	The animated camera follows Bat Neo returning back to the cave after a search for the balsa flower from 2:07am to 5:24am. This bat travels from mainland to the Isla Colón.



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Display of the 3D population density dataset for the Bocas del Toro Archipelago



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Concluding remarks (displaying the points flight paths, polygon land class, and line administrative boundary of Panama)

Fig. 1 - Storyboard

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Individual Reflections:

Ben Smith:

For Assignment 1, our group divided the work evenly. Instead of assigning specific project stages to individual members, we chose to collaborate on most sections, dividing up chapter work only after establishing key foundations such as data sources and stylistic choices. Before beginning the project, each group member proposed potential topics, which we then presented and discussed together. We collectively decided to pursue my idea of tracking animal migration, which was inspired by the boat tracking website Ian showed during a recent lecture. I thought it would be interesting to follow (ethically) an animal that would have more chaotic motion than a routed ship. From there, I searched the Movebank Data Repository (a site for peer-reviewed published animal tracking datasets) and found useful data called “Greater spear nosed bats commute long distances alone, rest together, but forage apart” in the form of a .csv file. This find narrowed down the scope of our project to this specific species in the specific region of Panama. As a result of my data find, I took on the responsibility of converting the .csv file into a series of points for 2 specific bat flight paths. This required file filtering and manipulation in ArcGIS Pro. Although working on the 3D dataset was assigned to the group as a whole, I was able to find the dataset for population density in Panama as well from the Humanitarian Data Exchange. This also required some editing in ArcGIS Pro before converting to a geoJSON was possible, including cropping

the area to the study region extent and converting the .tiff format to polygon features. As an overview of my personal coding responsibilities for the assignment, I took on the tasks of embedding a video and creating a list of buttons. For the video, I followed the advice of W3Schools (HTML YouTube Videos page) and incorporated the `<iframe>` element into the chapter description. The goal of the button list was to allow the user to click on a bat's name and view only its individual flight path, hiding all others on the map. To do this, I created three functions using `map.setLayoutProperty` (as used by the *show and hide layers* example on Mapbox GL JS). Besides these tasks, other responsibilities included working collaboratively on styling properties (such as fonts, alignment, colour and text size), adding images to the chapters, controlling the pitch/bearing for transitions, and writing chapter descriptions (specifically for the introduction and background chapters).

To tell the story of our two selected bats from Isla Colón, Panama, we began by creating a line GeoJSON of the national administrative boundary. This allowed us to clearly define the country of study for our audience. Next, we created a polygon GeoJSON to highlight the areas where the ephemeral balsa trees are found (a valuable nectar source for the bats). Therefore, by filtering for only the secondary mixed broadleaf forest land class from a panama land use dataset, we were able to emphasize the target locations for the bats during their respective flights. Bat flights were represented using point GeoJSON data to help the audience visualize the approximate three-hour flight paths in relation to the highlighted land types. Finally, the 3D GeoJSON was created to help the audience visualize the spatiotemporal relationship between the bats and humans. Personally, I was responsible for the flight path GeoJSON (points). I also was able to display the 3D GeoJSON by styling the layer as a fill extrusion.

As mentioned above, my original idea for the project was to map large scale migration patterns for animal species, much like how the website that Ian showed in class mapped boating positions globally (marinetraffic.com). However, due to the two week time frame and limited skills for the assignment, it was not feasible to track multiple species across large areas. Not only were there not enough easily accessible datasets, but I quickly realized that the [geoJSON.io](https://geojson.io) website could not support very large files. Since we did not have the time to filter these big datasets, we decided to go with a smaller study area so that we could really focus on detail and story, rather than spending most of the time on data analysis. Additionally, the dataset that I found for the flight paths initially had data for 18 individual bats (O'Mara & Dechmann, 2023). Although it would have been interesting to examine the similarities and differences between the individuals, the number of points were not supported by Mapbox. From a visual standpoint, the storymap also appeared cluttered and less aesthetically pleasing due to the high number of points, which risked confusing the audience. Furthermore, if time had permitted, I would have liked to include a time slider in the story map to provide temporal context for the flight paths. This would allow viewers to select a specific time on the slider and see where each individual bat was at that moment, creating a more spatiotemporal understanding of their movements. Inspiration from this was taken from ArcGIS StoryMaps Gallery page, specifically from the *Rising Waters* project that used buttons to allow the audience to see the rainfall accumulation on specific dates defined by the button label. Finally, if time permitted, I would have liked to spend more time on the visualization for the 3D dataset. Despite loading the 3D data into the style, we had to maintain a far distance zoom for the layer to load on the storymap. We were unable to resolve this issue, and therefore, I would have liked to investigate this problem further.

Regarding the interactivity of our story map, the dynamic spatial data enhanced the narrative by reflecting the movement of the bats. Since the central theme of our assignment was to visualize the flight paths of the Greater Spear-Nosed Bat, it was essential for the camera to not only flow to the starting point, but also follow the path chronologically. This can be seen in the flight path for Bat Neo, where we animated the camera along the path of time-stamped points. This approach was essential for truly understanding both the movement patterns and the individual journey of the bat. We also chose to rotate the map for some chapters, including chapter 3 where the Island of the cave habitat was identified and explained. This technique allowed for the audience to view the landmass from various angles. When the chapters focused more on data than on landscape, we chose not to rotate the screen to avoid confusion and unnecessary detail. In addition, keeping the mixed broadleaf forest land class polygons static on our website allows the audience to more easily observe the spatial patterns of bat flight in relation to the locations of the balsa trees.

Jack Ge:

In this assignment, I was responsible for finding the line polygon dataset, writing about the starting point of the bats being analyzed - specifically, the Cuevas y Santuario La Gruta nature reserve in Bocas del Toro, Panama. I also worked on the chapters which talked about the 3D polygons used in our storymap which describes human populations, as well as working on some of the code to implement the modifications of the chapters I was tasked to work on. This involved ensuring that the chapters I worked on appeared with the same style as the other ones present in the storymap.

For this assignment, we used various geoJSON datasets in order to create our storymap. Among them, we used a line dataset which represented the provincial borders of Panama, a point dataset which showcases the migration paths taken by the bats, a land cover dataset that showcases the forest cover and land uses of Isla Colón, Panamá, and a population dataset which we decided to represent it by 3D polygons to showcase the presence of human populations near the bats' habitat. For the overall style of our storymap, we decided to go with a pink text box on a gray background to make it more clear to those viewing the storymap, along with choosing different shades of pink and purple to represent the flight paths so it's clear to the viewer. We wanted to choose the colors carefully so that it remains accessible to everyone viewing our storymap and that the information is presented as clear as possible to the audience.

Originally, we had hoped to implement a time slider bar which was to be used to control the progression of the flights taken by the bats. In the end, this idea was dropped due to time constraints and complications from implementing it into our storyboard (Create a time slider, n.d.). Moreover, I had planned to find a digital elevation model dataset of Isla Colón, but there was limited data available online that could be used. Some of the data I found was even behind a paywall. The ones I did manage to find that were free were incompatible with Mapbox since it was only compatible with 8-bit datasets whereas the one I found was 16-bit (Upload data to Mapbox, n.d.). We realized that it was much harder than expected to find data we could use to showcase the terrain heights and for the one I did find, I realized that turning the dataset from 16-bit to 8-bit was no easy task and even when I did try to change it using some python codes, the end result dataset did not accurately represent the realities found in the real world in that the data indicated that there were plenty of large flat areas. Taking that into account, we decided it would be best to use a population density dataset as a 3D dataset to represent the distance of human populations to the habitat of the bats.

We had also thought about adding bats to our storymap that were going to be animated in a way that they were going to fly into a cave that was going to be created by means of 3D polygons. We quickly realized that this was going to be more complicated than we expected, especially with how difficult it was for us to find digital elevation model datasets for the area we were analyzing. Moreover, the coding that needed to be done to be able to make the bats seem like they were flying into the caves was going just as complicated, if not more than trying to implement a raster dataset of terrain levels. Ultimately in the end, we decided to use population to represent our 3D dataset, meaning that there wasn't going to be any point in implementing our idea of flying bats throughout the storymap.

The dynamic and static spatial data we used played a key part of our storymap in that it strengthened the narrative process of our story. The provincial borders of Panama dataset allowed us to augment the narrative process by providing geographic information about the setting of the storymap, while the point data provided migration information about the bats we were analyzing. Despite these features not being necessary, having some information provided visually to the audience allowed the reader to be further engaged, making the migration patterns easier to understand by the public. Not only did all of the elements present in the storymap provide a more clear picture of the bats, they also can provide a more interactive, immersive story map for the reader. Compared to having only text as sources of information, our storymap provides more visual details that allow for different ways of interpretation of information, allowing for the audience to form connections between human activity and their potential effects on the bat populations and their migration patterns.

Kyra Meier:

I was responsible for many aspects of this assignment, from the Mapbox style assembly to certain config.js edits and comments for the entire script, to the softer, conceptual storyflow of the chapters, as

well as the land cover polygons dataset including clipping, processing, and converting the shapefile to geojson, in addition to my spin-off Bat Simulator site which technical handling of point data and coding.

Some different geojson objects, tilesets, and Mapbox style customizations I used to set the story apart were effective and vibrant. A lot had to do with colour. I enjoyed customizing colour codes (for example, #ff79d9 pink for the lighter pink of the inset marker and the bat geo path) and creating a visually pleasing storyboard for the user to move through while enjoying what could have been a heavily scientific presentation. As for the Mapbox Style customization, I chose to style the main site group3/assignment1 map with varying bright colours and a semi-opaque tone for the 3D dataset. I also chose the mapbox standard satellite image for the basemap, rather than streetview, to emphasize the natural focus of this project - bats like trees, not roads! Another feature of the Mapbox styling I loved was to remove visually distracting labels and transit networks and to set the light preset to dawn, the time when greater spear-nosed bats make their journeys. After a bit of toggling, trial and error, and a quick learning curve of inputting the layers as tilesets (including accidentally deleting a fundamental tileset), I added our style url and access key to the HTML script, initiating our website.

Another customization technique I enjoyed employing was the Bat Simulator branch-out which takes the user from the main story map (assignment1) to another page (assignment1/batsimulator) through an embedded URL. This required its own tileset, access key, styling, etc, which I set up to match our main style quite closely. I only added one layer to this tile, points, but it took quite a bit of pre-processing and data handling. I had to take Ben's points layer (O'Mara et. al 2021) and manually isolate 1) a single bat's path (#74DE47) and 2) that bat's location data by the minute, whereas it was entered by the second. The second-based data was of too high volume for processing so I chose to keep it simple and take one coordinate per minute, a rewarding choice for the end bat-simulator user but one requiring much pre-processing. Automation would have been simpler but I didn't find myself with the time to teach myself the skill.

I created my bat simulator using a default Mapbox example, [Animate the Camera Along a Path](#) which gave me an html script with much critical code. I changed it by taking out the camera view (it caused glitches) and editing all lines to follow the line view simply. I of course customized the code to my points by adding my Mapbox style uniquely created for the simulator, manually adding the coordinates of bat readings per minute, and changing certain colours and other features including camera altitude from ground, zoom level, pitch, and bearing. I did not rely on many sources for this project, rather much trial and error. I was also inspired by the [Tenth Mountain Hut storyboard](#) from huttrip and inspected the source for inspiration.

The bat simulator view served as a replacement for a timeline feature, something I had been working for during much of the project and wanted to employ to showcase the high quality of the data - unique coordinate readings per second for each bat in the study, collected through attached trackers (O'Mara et. al. 2023). I had planned to create a timeline to toggle where the bat was at a certain time compared to the next. I finally gave up on this idea when I realized every example I looked at referred to creating a timeline for the entire story, not just one chapter like I wanted to do. Although my lack of expertise kept me from realizing this vision, I am grateful to it for directing me in a new, even more fun way - the bat simulator.

Sometimes storytelling doesn't actually need maps or interactivity but they're there anyway. However, that was not the case for our story. The dynamic and static spatial data augment the narrative process very strongly by allowing genuine visualization of a spatial phenomenon, the resource-based migration of Spear-nosed bats. With our spatial data, the user is able to understand how bat flight paths, Neo and Geo, overlap with relevant land cover classes in search of the Balsa flower, while overlapping and avoiding population concentrations in an interesting way. By actually visualizing this journey, we understand how important it is to protect these bats' precious feeding grounds like at the Cuevas y Santuario La Gruta nature reserve, because it is the only one within miles and bats fly for hours to reach it (O'Mara et. al 2023). I think this narrative could be told in other ways, but it would have to be thoroughly explained and would leave a lot up to interpretation of the viewer, especially a government clientele who needs clear data in order to make political decisions. The dynamic and static spatial data allows for clear visualization of the problem, its place in the world, and the scope of its importance.